

Logic / Symbolic Logic:

Symbolic logic may be thought of as a formal language for representing facts about objects and relationships between objects of a problem domain along with a precise inferencing mechanism for reasoning and deduction. An **inferencing mechanism** derives the knowledge, which is not explicitly/directly available in the knowledge base, but can be logically inferred from what is given in the knowledge base.

In the **propositional logic**, we are interested in declarative sentences, i.e., sentences that can be either true or false, but not both. Any such declarative sentence is called a **proposition or a statement**. For example

- (i) The proposition: “The sunrises in the west,” is False,
- (ii) The proposition: “Sugar is sweet,” is True, and
- (iii) The truth of the proposition: “Ram has a Ph. D degree.” depends upon whether Ram is actually a Ph. D or not.

For a given declarative sentence, its being ‘True’ or ‘False’ is called its **Truth-value**.

On the other hand, none of the following sentences can be assigned a truth-value, and hence none of these, is a statement or a proposition:

- (i) Who was the first Prime Minister of India? (Interrogative sentence)
- (ii) Please, give me that book. (Imperative sentence)
- (iii) Ram must exercise regularly. (Imperative, rather Deontic)
- (iv) Hurrah! We have won the trophy. (Exclamatory sentence)

In propositional logic, as mentioned earlier also, symbols are used to denote propositions. For instance, we may denote the propositions discussed above as follows:

- P : The sunrises in the west,
- Q : Sugar is sweet,
- R : Ram has a Ph.D. degree.

The symbols, such as P, Q, and R, that are used to denote propositions, are called **atomic formulas, or atoms**. As discussed earlier, in this case the truth-value of P is False, the truth-value of Q is True and the truth-value of R, though not known yet, is exactly one of ‘True’ or ‘False’, depending on whether Ram is actually a Ph. D or not.

At this stage, it may be noted that once symbols are used in place of given statements in, say, English, then the propositional system, and, in general, a symbolic system is aware **only** of symbolic representations, and the associated truth values. The system operators only on these representations. And, except for possible final translation, is **not aware** of the originally statements, generally given in some natural language, say, English.

We can build, from atoms, more complex propositions, sometimes called **compound propositions**, by using logical connectives.

Examples of such propositions are:

- (i) Sunrises in the east **and** the sky is clear, and
- (ii) **If** it is hot **then** it shall rain.

The logical connectives in the above two propositions are “and” and “if...then”. In the propositional logic, **five logical operators or connectives**, viz., $\sim$ (not), $\wedge$ (and), $\vee$ (or), $\rightarrow$ (if... then), and $\leftrightarrow$ (if and only if), are used. These five logical connectives can be used to build

compound propositions from given atomic formulas. More generally, they can be used to construct more complicated compound propositions from compound propositions by applying the connectives repeatedly. For example, **if** each of the letters P, Q, C is used as a symbol for the corresponding statement, as follows:

P: The wind speed is high.

Q: Temperature is low.

C: One feels comfortable.

then the sentence:

“If the wind speed is high and the temperature is low, then one does not feel comfortable” may be represented by the formula $((P \wedge Q) \rightarrow (\sim C))$. Thus, a compound proposition can express a complex idea. In the propositional logic, an expression that represents a proposition, such as P, or a compound proposition, such as $((P \wedge Q) \rightarrow (\sim C))$, is called a well-formed formula.

7.1.1 Logic & Valid Arguments

Logic is the analysis and appraisal of arguments.

An **argument** is a set of statements consisting of a finite number of premises, i.e., assumed statements and a conclusion.

Valid Argument: A valid argument is one in which it would be contradictory for the premises to be true but the conclusion false.

Example of Valid Argument

(i) If you overslept, you will be late

(ii) You are not late.

Means you did not over sleep.

Example of Invalid argument

(i) If you overslept, you will be late

(ii) You did not oversleep

Means you are not late

(This argument is invalid, because despite not having overslept, one may be late because of some other engagements or laziness.)

Another Invalid Argument

(i) If we are close to the top of Mt. Everest then we have magnificent view.

(ii) We are having a magnificent view.

Therefore,

(iii) We are the near the top of Mt. Everest.

(This argument is invalid, because, we may have a magnificent view even if we are not close to the top of Mt. Everest. The two given statements do not falsify this claim)

Another Invalid Argument

(i). If you are in Delhi, you are in India.

(ii). You are in India.

(iii). Therefore, you are in Delhi (*invalid argument, though conclusion may be True*)

7.1.2 Non Logical Operators: One of reason why special **symbols**:

$\wedge \vee \sim \leftrightarrow \rightarrow$

are used in symbolic logic instead of the corresponding natural languages **words**: and or not if... then, if and only if, is that the **words** may have different meaning in different contexts. For example, the use of the word **and** in one sentences has different connotations or meanings from the use in others in the following:

(i) Ram **and** Mohan are good hockey players.

(the statement can be equivalently broken into two statements:

(i) Ram is a good hockey player (ii) Mohan is a good hockey player)

(ii) Ram **and** Mohan are good friends.

(though the word **and** joins two words Ram & Mohan, but can not be equivalently broken into two statements)

(iii) Mohan drove a car to reach home, met an accident and got slightly injured.

(Here, the use of the word and is not in a logical sense, but, it is in temporal sense of and then because statement (iii) has different sense from the statement given in (iv) below)

(iv) Mohan met an accident, got slightly injured **and** drove a car to reach home.

Thus from the above statements, it can be seen that the natural language word **and** may have many senses, both logical and non-logical. Similarly, the **words since, hence and because** are frequently used in arguments to establish some facts. But as shown from the following two arguments, their use in logical arguments is **risky** in the sense that some of the arguments involving any of these words may lead to **incorrect** conclusions:

7.1.3 Propositional Logic

A Well-formed formula, or wff or formula in short, in the propositional logic defined recursively as follows:

1. An atom is a wff.
2. If A is a wff, then $(\sim A)$ is a wff.
3. If A and B are wffs, then each of $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, and $(A \leftrightarrow B)$ is a wff.
4. Any wff is obtained only by applying the above rules.

From the above recursive definition of a wff it is not difficult to see that expression:

$((P \rightarrow (Q \wedge (\sim R))) \leftrightarrow S)$ is a wff; because , to begin with, each of P, Q , $(\sim R)$ and S, by definitions is a wff. Then, by recursive application, the expression: $(Q \wedge (\sim R))$ is a wff. Again, by another recursive application, the expression: $(P \rightarrow (Q \wedge (\sim R)))$ is a wff. And, finally the expression given initially is a wff.

Further, it is easy to see that according to the recursive definition of a wff, each of the expressions: $(P \rightarrow (Q \wedge))$ and $(P (Q \wedge R))$ is **not** a wff.

Some pairs of parentheses may be dropped, for simplification. For example,

$A \vee B$ and $A \rightarrow B$ respectively may be used instead of the given wffs $(A \vee B)$ and $(A \rightarrow B)$, respectively. We can omit the use of parentheses by assigning priorities in increasing order to the connectives as follows:

$\leftrightarrow, \rightarrow, \wedge, \vee, \sim$.

Thus, ' $\leftrightarrow$ ' has least priority and ' $\sim$ ' has highest priority. Further, if in an expression, there are no parentheses and two connectives between three atomic formulas are used, then the operator with higher priority will be applied first and the other operator will be applied later.

For example: Let us be given the wff $P \rightarrow Q \wedge \sim R$ without parenthesis. Then among the operators appearing in wff, the operator ' $\sim$ ' has highest priority. Therefore, $\sim R$ is replaced by $(\sim R)$. The equivalent expression becomes $P \rightarrow Q \wedge (\sim R)$. Next, out of the two operators viz ' $\rightarrow$ ' and ' $\wedge$ ', the operators ' $\wedge$ ' has higher priority. Therefore, by applying parentheses appropriately, the new expression becomes $P \rightarrow (Q \wedge (\sim R))$. Finally, only one operator is left. Hence the fully parenthesized expression becomes $(P \rightarrow (Q \wedge (\sim R)))$

Now, we define the rules of finding the truth value or meaning of a wff, when truth values of the atoms appearing in the wff are known or given.

- 1) The wff $\sim A$ is True when A is False, and $\sim A$ is False when A is true. The wff $\sim A$ is called the **negation** of A.
- 2) The wff $(A \wedge B)$ is True if A and B are both True; otherwise, the wff $A \wedge B$ is False. The wff $(A \wedge B)$ is called the **conjunction** of A and B.
- 3) The wff $(A \vee B)$ is true if at least one of A and B is True; otherwise, $(A \vee B)$ is False. $(A \vee B)$ is called the **disjunction** of A and B.
- 4) The wff $(A \rightarrow B)$ is False if A is True and B is False; otherwise, $(A \rightarrow B)$ is True. The wff $(A \rightarrow B)$ is read as "If A, then B," or "**A implies B.**" The symbol ' $\rightarrow$ ' is called **implication**.
- 5) The wff $(A \leftrightarrow B)$ is True whenever A and B have the same truth values; otherwise $(A \leftrightarrow B)$ is False. The wff $(A \leftrightarrow B)$ is read as "**A if and only if B.**"

Table 7.2: Truth table

A	B	$\sim A$	$(A \wedge B)$	$(A \vee B)$	$(A \rightarrow B)$	$(A \leftrightarrow B)$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Validity through Truth-Table.

- (i) If I overslept, then I am late, i.e., symbolically

$$S \rightarrow L$$

- (ii) I am not late, i.e., symbolically

$$\sim L$$

To conclude

- (iii) I did not oversleep, i.e., symbolically

$$\sim S$$

To establish the validity/Invalidity of the argument, consider the Truth-Table

S	L	$S \rightarrow L$	$\sim L$	$\sim S$
F	F	T	T	T
F	T	T	F	T
T	F	F	T	F
T	T	T	F	F

There is only one row, viz., first row, in which both the premises viz. $S \rightarrow L$ and $\sim L$ are True. But in this case the conclusion represented by $\sim S$ is also True. Hence, the conclusion is valid.

Example 7.1: Express the following statements in Propositional Logic.

- a) If he campaigns hard, he will be elected.
- b) If the humidity is high, it will rain either today or tomorrow.
- c) Cancer will not be cured unless its cause is determined & a new drug for cancer is found.
- d) It requires courage and skills to climb a mountain.

Solution:

- (a) Let H: He campaigns hard ; E: He will be elected

Then the statement becomes the formula:

$$H \rightarrow E$$

- (b) Let H: The Humidity is high, RTY: It will rain today

RTW: It will rain tomorrow.

Then

$$H \rightarrow RTY \vee RTW$$

- (c) Let C: Cancer will be cured

D: Cancer's cause will be determined

F: A new drug for cancer will be found

Then the statement becomes the formula:

$$(\sim C) \vee (D \wedge F). \text{ This formula may also be written as:}$$

$$C \rightarrow D \wedge F$$

- (d) Let C: One has courage

S: One has skill

M: One climbs mountain

Then the statement becomes the formula:

$$M \rightarrow C \wedge S$$

Example 7.2: Let

P : He needs a doctor,

Q : He needs a lawyer,

R : He has an accident,

S : He is sick,

U : He is injured.

State the following formulas in English.

a) $(S \rightarrow P) \wedge (R \rightarrow Q)$

b) $P \rightarrow (S \vee U)$

c) $(P \wedge Q) \rightarrow R$

d) $(P \wedge Q) \leftrightarrow (S \wedge U)$

Solution:

(a) If he is sick then he needs a doctor, but, if he has an accident then he needs a lawyer

(b) If one requires a doctor then one must be either sick or injured.

(c) If he needs both a doctor and a lawyer then he has an accident.

(d) One requires a doctor and also a lawyer if and only if one is sick and also injured.

Example 7.3: Express the following sentences into propositional form.

1) If is not raining and I have time, then I will go to a Movie.

2) If it is raining and I will not go to a movie.

3) It is not raining.

4) I will not go to a movie.

5) I will go to a movie only if it is not raining.

Solution: Consider the following assumptions.

Let P represent the predicate 'it is raining'

Q represent the predicate 'I have the time'

R represent the predicate 'I will go to Movie'

Then the above statements can be written as

- 1) $(\sim P \wedge Q) \rightarrow R$
- 2) $P \wedge \sim R$
- 3) $\sim P$
- 4) $\sim R$
- 5) $R \rightarrow \sim P$

Example 7.4: If P, Q, R the propositions defined as

P represent the predicate 'it is raining'

Q represent the predicate 'I have the time'

R represent the predicate 'I will go to Movie'

Write the sentence in English corresponding to the following proposition form.

- 1) $(\sim P \wedge Q) \leftrightarrow R$
- 2) $(Q \rightarrow R) \wedge (R \rightarrow Q)$
- 3) $\sim(Q \vee R)$
- 4) $R \rightarrow \sim P \wedge Q$

Solution:

- 1) I will go to a movie, if it is not raining and I will have the time.
- 2) I will go to a movie, if and only if I have the time.
- 3) It is not the case that I have time or I will go to a movie.
- 4) I will go to a movie if it is not raining or I have time.

7.4.5 Interpretations of Formulas

In order to find the truth value of a given formula G, the truth values for the atoms of the formula are either given or assumed. The set of initially given/assumed values of all the atomic formulas occurring in a formula say G, is called an **interpretation** of the formula G. By developing a Truth-table of any formula, its truth value can be evaluated in terms of its interpretation, i.e., in terms of the truth values associated with the constituent atoms.

Consider the formula

$G : ((A \wedge B) \rightarrow (R \leftrightarrow (\sim S)))$. G, before the symbol ':', is the name of the formula.

The assignment of the truth values T, F, T, T to A, B, R, S, respectively, is called an **interpretation of the formula G**. Since, each one of A, B, R, and S can be assigned one of the two values, viz., either T or F, there are $2^4 = 16$ possible interpretations of the formula G. In Table 7.3, we give the truth values of the formula G under all these 16 interpretations.

The above procedure may be repeated to find truth value of any formula from any interpretation, i.e., from any assignment to the atomic formulas occurring in the given formula.

Table 7.3: Truth Table of $(A \wedge B \rightarrow (R \leftrightarrow (\sim S)))$

A	B	R	S	$\sim S$	$(A \wedge B)$	$(R \leftrightarrow (\sim S))$	$(A \wedge B) \rightarrow (R \leftrightarrow (\sim S))$
T	T	T	T	F	T	F	F
T	T	T	F	T	T	T	T
T	T	F	T	F	T	T	T
T	T	F	F	T	T	F	F
T	F	T	T	F	F	F	T
T	F	T	F	T	F	T	T

T	F	F	T	F	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	F	F	F	T
F	T	T	F	T	F	T	T
F	T	F	T	F	F	T	T
F	T	F	F	T	F	F	T
F	F	T	T	F	F	F	T
F	F	T	F	T	F	T	T
F	F	F	T	F	F	T	T
F	F	F	F	T	F	F	T

A table, such as given above, that displays the truth values of a formula G for all possible assignments of truth values to atoms occurring in G is called a **Truth table of G** .

NOTATION: If $A_1, \dots, A_n$ are all the atoms in a formula, it may be more convenient to represent an *interpretation* by a set $(m_1, \dots, m_n)$, where m_i is either A_i or $\sim A_i$. m_i is written as A_i if T is assigned to A_i . But m_i is written as $\sim A_i$ if F is assigned to A_i .

The set $\{A, \sim B, \sim R, S\}$ represents an interpretation of a formula in which A, B, R , and S are the only atoms and which are, respectively, assigned T, F, F, and T. We will use the notation throughout.

Example 7.5: Construct a truth table for the formula.

$$P: (\sim A \vee B) \wedge (\sim (A \wedge \sim B))$$

Solution:

Table 7.4: Truth table for $(\sim A \vee B) \wedge (\sim (A \wedge \sim B))$

A	B	$\sim A$	$\sim B$	$\sim A \vee B$	$A \wedge \sim B$	$\sim (A \wedge \sim B)$	P
T	T	F	F	T	F	T	T
T	F	F	T	F	T	F	F
F	T	T	F	T	F	T	T
F	F	T	T	T	F	T	T

7.4.6 Validity and inconsistency of Propositions

we shall consider wff that are **true under all** possible interpretations and wff that are **false under all** possible interpretations.

For example Let us consider the wff

$$G: (((A \rightarrow B) \wedge A) \rightarrow B).$$

The formula G has $2^2 = 4$ possible interpretations in view of the fact it has two atoms viz A and B . It can be easily seen from the following table that the wff G is True under all its interpretations. **Such as a wff which is True under all interpretation is called a valid formula (or a tautology).**

Table 7.5: Truth Table of $(((A \rightarrow B) \wedge A) \rightarrow B)$

A	B	$(A \rightarrow B)$	$(A \rightarrow B) \wedge A$	$(((A \rightarrow B) \wedge A) \rightarrow B)$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Consider another formula

$$G : ((A \rightarrow B) \wedge (A \wedge \sim B))$$

The truth table of the formula G given below shows that G is False under all its interpretations. Such a formula which is False under all interpretations is called an **inconsistent formula (or a contradiction)**.

Table 7.6: Truth Table of $(A \rightarrow B) \wedge (A \wedge \sim B)$

A	B	$\sim B$	$(A \rightarrow B)$	$(A \wedge \sim B)$	$((A \rightarrow B) \wedge (A \wedge \sim B))$
T	T	F	T	F	F
T	F	T	F	T	F
F	T	F	T	F	F
F	F	T	T	F	F

Next, we formally define the concepts discussed above.

Definition: A formula is said to be **valid** if and only if it is **true under all** its interpretations. A formula is said to be **invalid** if and only if it is *not true under at least one* interpretation. A valid formula is also called a **Tautology**. A formula is **invalid** if there is **at least one** interpretation for which the formula has a truth value False.

Definition: A formula is said to be **inconsistent (or unsatisfiable)** if and only if it is **False under all** its interpretations. A formula is said to be **consistent or satisfiable** if and only if it is not inconsistent. In other words, a formula is *consistent if there is at least one interpretation* for which the formula has a truth value true.

From the definitions given above, it is easily seen that

- (i) A formula is *valid* if and only if its negation is *inconsistent*.
- (ii) A formula is *invalid* if and only if there is *at least one* interpretation under which the formula is *false*.
- (iii) A formula is *consistent* if and only if there is *at least one* interpretation under which the formula is *true*.
- (iv) If a formula is *valid*, then it is *consistent*, but not *vice versa*. (example given below)
- (v) If a formula is *inconsistent*, then it is *invalid*, but not *vice versa*. (example given below)

Example 7.6: For each of the following formulas, determine whether it is valid, inconsistent, consistent or some combination of these.

- (i) E: $\sim (\sim A) \rightarrow B$
- (ii) G: $(A \rightarrow B) \rightarrow (\sim B \rightarrow \sim A)$
- (iii) H: $(A \vee \sim A) \rightarrow (A \wedge B) \wedge (\sim A)$
- (iv) J: $(A \wedge B) \wedge (\sim A) \rightarrow (B \vee \sim B)$

Solution:

- (i) Consistent but not valid, because, for For B as T and A as F, the formula is T. But, for A as T and B as F the formula is F.

- (ii) It can be easily that $\sim B \rightarrow \sim A$ has same truth-value as $(A \rightarrow B)$ for any interpretation. Therefore, instead of given formula, we may consider the formula $(A \rightarrow B) \rightarrow (A \rightarrow B)$ which can be further written as $P \rightarrow P$, writing $(A \rightarrow B)$ as P . Even $P \rightarrow$ can be written as $P \vee P \equiv P \equiv (A \rightarrow B)$, The last formula is F when F and A is T. The formula is T when A is F and B is T. Hence, the formula is neither valid nor inconsistent.
Therefore, the formula is consistent but not valid
- (i) For all truth assignments to A and B , L. H.S. of the formula is always T and R. H.S. is always F. Hence the formula is inconsistent, i.e., always F
- (ii) The L. H. S. of the given formula is F under all interpretations. Hence, the formula is T under all interpretation. Therefore, the formula is valid.

7.4.7 Equivalent Forms in the Propositional Logic (PL)

Definition: Logically Equivalent Formulas: Two formulas G_1 and G_2 are said to be (logically) *equivalent* if for each interpretation i.e., truth assignment to all the atoms that occur in either G_1 or G_2 ; the truth values of G_1 and G_2 are identical. In other words, for each interpretation, G_1 is True if and only if G_2 is True. And, for each interpretation, G_1 is False if and only if G_2 is False.

As will be clear later, it is often necessary to transform a formula from one form to another, especially to a *normal form*. This is accomplished by replacing a formula in the given formula by a formula *equivalent* to it and repeating this process until the desired form is obtained.

verify that the formula E: $\sim (A \rightarrow B)$ is equivalent the formula G: to $A \wedge \sim B$ by examining the following truth table.

Table 7.7: Table Joint Truth table of $\sim (A \rightarrow B)$ and $(A \wedge \sim B)$

A	B	$\sim B$	$(A \rightarrow B)$	$\sim(A \rightarrow B)$	$A \wedge \sim B$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

The corresponding values in the last two columns are identical, so both are equivalent.

Solutions of problems using symbolic logic can be simplified, if we can simplify involved formulas by some equivalent simpler formulas given in table below. These equivalences can be verified by using truth tables.

Table 7.8: Equivalences of PL

(1.1)	$E \leftrightarrow G = (E \rightarrow G) \wedge (G \rightarrow E)$	
(1.2)	$E \rightarrow G = \sim E \vee G$	
(1.3)(a)	$E \vee G = G \vee E;$	(b) $E \wedge G = G \wedge E$
(1.4)(a)	$(E \vee G) \vee H = E \vee (G \vee H);$	(b) $(E \wedge G) \wedge H = E \wedge (G \wedge H)$
(1.5)(a)	$E \vee (G \wedge H) = (E \vee G) \wedge (E \vee H);$	(b) $E \wedge (G \vee H) = (E \wedge G) \vee (E \wedge H)$
(1.6)(a)	$E \vee \text{False} = E;$	(b) $E \wedge \text{True} = E$
(1.7)(a)	$E \vee \text{True} = \text{True}$	(b) $E \wedge \text{False} = \text{False}$
(1.8)(a)	$E \vee \sim E = \text{True};$	(b) $E \wedge E = E$
(1.9)	$\sim(\sim E) = E$	

$$(1.10)(a) \quad \sim (E \vee G) = \sim E \wedge \sim G;$$

$$(b) \quad \sim (E \wedge G) = \sim E \vee \sim G$$

In the table given above, True denotes the fact that the wff is True under all interpretations and False denotes the wff that is False under all interpretations.

Laws (1.3a), (1.3b) are often, called **commutative laws**; (1.4a), (1.4b) **associative laws**; (1.5a), (1.5b), **distributive laws**; and (1.10a), (1.10b), **De Morgan's laws**.

7.5 NORMAL FORMS

A **clause** is a disjunction of literals. For example, $(E \vee \sim F \vee \sim G)$ is a clause. But $(E \vee \sim F \wedge \sim G)$ is not a clause. A **literal** is an atom, say A, or its negation, say $\sim A$.

Definition: A formula E is said to be in a **Conjunctive Normal Form (CNF)** if and only if E has the form $E : E_1 \wedge \dots \wedge E_n$, $n \geq 1$, where each of $E_1, \dots, E_n$ is a **disjunction** of literals.

Definition: A formula E is said to be in **Disjunctive Normal Form (DNF)** if and only if E has the form $E : E_1 \vee E_2 \vee \dots \vee E_n$, where each E_i is a **conjunction** of literals.

Note: Let A, B and C be atoms. Then $F : (\sim A \wedge B) \vee (A \wedge \sim B \wedge \sim C)$ is a formula in a disjunctive normal form. Again $G : (\sim A \vee B) \wedge (A \vee \sim B \vee \sim C)$ is a formula in Conjunctive Normal Form, because it is a conjunction of the two disjunctions of literals viz of $(\sim A \vee B)$ and $(A \vee \sim B \vee \sim C)$

Example 7.7: Each of the following is neither in CNF nor in DNF

$$(i) \quad (\sim A \vee B) \vee (A \wedge \sim B \vee C)$$

$$(ii) \quad (A \rightarrow B) \wedge (\sim B \wedge \sim A)$$

Using table of equivalent formulas given above, any valid Propositional Logic formula can be transformed into CNF as well as DNF.

Solution: The steps for conversion to DNF are as follows

Step 1: Use the equivalences to remove the logical operators ' $\leftrightarrow$ ' and ' $\rightarrow$ ':

$$(i) \quad E \leftrightarrow G = (E \rightarrow G) \wedge (G \rightarrow E)$$

$$(ii) \quad E \rightarrow G = \sim E \vee G$$

Step 2 Remove $\sim$'s, if occur consecutively more than once, using

$$(iii) \quad \sim(\sim E) = E$$

(iv) Use De Morgan's laws to take ' $\sim$ ' nearest to atoms

$$(v) \quad \sim(E \vee G) = \sim E \wedge \sim G$$

$$(vi) \quad \sim(E \wedge G) = \sim E \vee \sim G$$

Step 3 Use the distributive laws repeatedly

$$(vii) \quad E \vee (G \wedge H) = (E \vee G) \wedge (E \vee H)$$

$$(viii) \quad E \wedge (G \vee H) = (E \wedge G) \vee (E \wedge H)$$

Example 7.8: Obtain a disjunctive normal form for the formula $\sim(A \rightarrow (\sim B \wedge C))$.

Solution:

$$\begin{aligned}
\text{Consider } A \rightarrow (\sim B \wedge C) &= \sim A \vee (\sim B \wedge C) && \text{(Using } (E \rightarrow F) = (\sim E \vee F) \text{)} \\
\text{Hence, } \sim (A \rightarrow (\sim B \wedge C)) &= \sim (\sim A \vee (\sim B \wedge C)) \\
&= \sim (\sim A) \wedge (\sim (\sim B \wedge C)) && \text{(Using } \sim (E \vee F) = \sim E \wedge \sim F \text{)} \\
&= A \wedge (B \vee (\sim C)) && \text{(Using } \sim (\sim E) = E \text{ and } \sim (E \wedge F) = \sim E \vee \sim F \text{)} \\
&= (A \wedge B) \vee (A \wedge (\sim C)) && \text{(Using } E \wedge (F \vee G) = (E \wedge F) \vee (E \wedge G) \text{)}
\end{aligned}$$

However, if we are to obtain CNF of $\sim A \rightarrow (\sim B \wedge C)$, in the last but one step, we obtain $\sim (A \rightarrow (\sim B \wedge C)) = A \wedge (B \vee \sim C)$, which is in CNF, because, each of A and $(B \vee \sim C)$ is a disjunct.

Example 7.9: Obtain conjunctive Normal Form (CNF) for the formula: $D \rightarrow (A \rightarrow (B \wedge C))$

Consider

$$\begin{aligned}
D \rightarrow (A \rightarrow (B \wedge C)) &\quad \text{(using } E \rightarrow F = \sim E \vee F \text{ for the inner implication)} \\
= D \rightarrow (\sim A \vee (B \wedge C)) &\quad \text{(using } E \rightarrow F = \sim E \vee F \text{ for the outer implication)} \\
= \sim D \vee (\sim A \vee (B \wedge C)) \\
= (\sim D \vee \sim A) \vee (B \wedge C) &\quad \text{(using Associative law for disjunction)} \\
= ((\sim D \vee \sim A \vee B) \wedge (\sim D \vee \sim A \vee C))
\end{aligned}$$

The last line denotes the conjunctive Normal Form of $D \rightarrow (A \rightarrow (B \wedge C))$
(using distributivity of $\vee$ over $\wedge$)

Note: If we stop at the last but one stop, then we obtain $(\sim D \vee \sim A) \vee (B \wedge C) = \sim D \vee \sim A \vee (B \wedge C)$ is a Disjunctive Normal Form for the given formula: $D \rightarrow (A \rightarrow (B \wedge C))$.

Example 7.10: Transform the following into disjunctive normal forms.

$$(i). \sim (A \vee \sim B) \wedge (S \rightarrow T) \quad (ii). (A \rightarrow B) \rightarrow R$$

Solution:

$$\begin{aligned}
(i) \quad &\text{Removing '}\rightarrow\text{' , we get} \\
&\sim (A \vee \sim B) \wedge (\sim S \vee T) \\
&\text{Taking '}\sim\text{' inside we get} \\
&(\sim A \wedge B) \wedge (\sim S \vee T) \text{ (using De Morgan's Law)} \\
&\text{Using distributivity of } \wedge \text{ over } \vee \text{ we get} \\
&(\sim A \wedge B \wedge \sim S) \vee (\sim A \wedge B \wedge T) \\
&\text{which is the required form}
\end{aligned}$$

$$\begin{aligned}
(ii) \quad &\text{Removing outer } \rightarrow \text{ we get} \\
&\sim (A \rightarrow B) \vee R \\
&\text{Removing the other '}\rightarrow\text{' we get} \\
&\sim (\sim A \vee B) \vee R \\
&\text{Taking } \sim \text{ inside, we get} \\
&(A \wedge \sim B) \vee R, \\
&\text{which is the required form}
\end{aligned}$$

7.6 First Order Predicate Logic (FOPL):

In the previous part, we discussed how propositional logic helps us in solving problems. However, one of the major problems with propositional logic is that, sometimes, it is unable to capture even elementary type of reasoning or argument as represented by the following statements:

Every man is mortal.
Raman is a man.
Hence, he is mortal.

The above reasoning is intuitively correct. However, if we attempt to simulate the reasoning through Propositional Logic and further, for this purpose, we use symbols P, Q and R to denote the statements given above as:

P: Every man is mortal,
Q: Raman is a man,
R: Raman is mortal.

Once, the statements in the argument in English are symbolized to apply tools of propositional logic, we just have three symbols P, Q and R available with us and apparently no link or connection to the original statements or to each other. The connections, which would have helped in solving the problem, become invisible. In Propositional Logic, there is no way, to conclude the symbol R from the symbols P and Q. However, as we mentioned earlier, even in a natural language, the conclusion of the statement denoted by R from the statements denoted by P and Q is obvious. Therefore, we search for some **symbolic** system of reasoning that helps us in discussing argument forms of the above-mentioned type, in addition to those forms which can be discussed within the framework of propositional logic. **First Order Predicate Logic (FOPL)** is the most well-known symbolic system for the purpose.

In addition to the baggage of concepts of propositional logic, FOPL has the following additional concepts: terms, predicates and quantifiers. FOPL uses two special symbol $\forall$ and $\exists$ are used to denote qualifications. The symbols $\forall$ and $\exists$ are called, respectively, the universal quantifier and existential quantifier. For a variable x, $(\forall x)$ is read as for all x, and $(\exists x)$ is read as there exists an x.

In order to symbolize the following statements:

- i) There exists a number that is rational.
- ii) Every rational number is a real number
- iii) For every number x, there exists a number y, which is greater than x.

let us denote x is a rational number by $Q(x)$, x is a real number by $R(x)$, and x is less than y by $LESS(x, y)$. Then the above statements may be symbolized respectively, as

- (i) $(\exists x) Q(x)$
- (ii) $(\forall x) (Q(x) \rightarrow R(x))$
- (iii) $(\forall x) (\exists y) LESS(x, y)$.

Each of the expressions (i), (ii), and (iii) is called a **formula** or a well-formed formula or **wff**.

Example 7.11: Translate the text “Every man is mortal, john is a man, therefore john is mortal” in the FOPL statements.

Solution: As discussed earlier, let us denote

$\text{Man}(x) - x$ is a man
 $\text{Mortal}(x) - x$ is mortal
 Then “every man is mortal” can be represented by
 $(\forall x) (\text{MAN}(x) \rightarrow \text{MORTAL}(x))$,
 “Raman is a man” by
 $\text{MORTAL}(\text{Raman})$.

The whole argument can now be represented by
 $(\forall x) (\text{MAN}(x) \rightarrow \text{MORTAL}(x)) \wedge \text{MAN}(\text{Roman}) \rightarrow \text{MORTAL}(\text{Roman})$.

Example 7.12: Formulate FOPL of the following statements.

- 1) Every Natural number is an integer.
- 2) There exists a number that is an even number.
- 3) For every number x there exists a number y such that $x > y$.
- 4) Every woman loves a child.

Let us define – $\text{Even}(x)$, $\text{Natural}(x)$, $\text{Integer}(x)$, $\text{Less}(x, y)$, $\text{Woman}(x)$, $\text{Child}(x)$, $\text{Love}(x)$

Solution:

- 1) $\forall x \text{ Number}(x) \rightarrow \text{Integer}(x)$
- 2) $\exists x \text{ Even}(x)$
- 3) $\forall x \exists y \text{ Less}(x, y)$
- 4) $\forall x \text{ Woman}(x) \rightarrow \exists y \text{ Child}(y) \wedge \text{Love}(x, y)$

Example 7.13: Let $P(x)$ and $Q(x)$ represent “ x is a rational number” and “ x is a real number,” respectively. Symbolize the following sentences:

- (i) Every rational number is a real number.
- (ii) Some real numbers are rational numbers.
- (iii) Not every real number is a rational number.

Solution:

- (i) $(\forall x) (P(x) \rightarrow Q(x))$
- (ii) $(\exists x) (P(x) \wedge Q(x))$
- (iii) $\sim (\forall x) (Q(x) \rightarrow P(x))$

Example 7.14: Let $C(x)$ mean “ x is a used-car dealer,” and $H(x)$ mean “ x is honest.” Translate each of the following into English:

- (i) $(\exists x) C(x)$
- (ii) $(\exists x) H(x)$
- (iii) $(\forall x) C(x) \rightarrow \sim H(x)$
- (iv) $(\exists x) (C(x) \wedge H(x))$
- (v) $(\exists x) (H(x) \rightarrow C(x))$.

Solution:

- (i) There is (at least) one (person) who is a used-car dealer.
- (ii) There is (at least) one (person) who is honest.
- (iii) All used-car dealers are dishonest.
- (iv) (At least) one used-car dealer is honest.
- (v) There is at least one thing in the universe, (for which it can be said that) if that something is Honest then that something is a used-car dealer

Example 7.15: Express the following sentences involving predicates in symbolic form.

- Solution:** define

- 1) $\forall x \ C(x)$
- 2) $\exists x \ \sim S(x)$
- 3) $\forall x \ (C(x) \rightarrow S(x))$
- 4) $\exists x \ (S(x) \wedge \sim C(x))$
- 5) $\exists x \ (C(x) \wedge S(x))$

7.7 Resolution Method

The key idea for the resolution method is that it is a refutation proof -- use the knowledge base and the negated goal to obtain null clause (which indicates contradiction). Since the knowledge base by itself is consistent, the contradiction must be introduced by the negated goal. As a result, we have to conclude that the original goal is true. The resolution proof can be considered as a generalization of modus ponens, because it can do more things than modus ponens does. The knowledge base used in resolution proof takes the so called clause form, which is a form of conjunction of disjunctions: each individual clause is expressed with disjunction as the connective and no conjunction is allowed within each disjunction. For example, $(a \wedge b) \vee c$ is not in clause form, but it can be converted to clause form, because it can be rewritten as $(a \vee c) \wedge (b \vee c)$, which is in clause form, because both $(a \vee c)$ and $(b \vee c)$ are disjuncts. The advantage of restricting to clause form is to make the proof process as an automated one.

Resolution algorithm

Input: A set of clauses, called axioms and a goal.

Output: To test whether the goal is derivable from the axioms.

Begin

1. Construct a set S of axioms plus the negated goal.
2. Represent each element of S into conjunctive normal form (CNF) by the following steps:
 - a) Replace 'if-then' operator by NEGATION and OR operation.
 - b) Bring each modified clause into the following form and then drop AND operators connected between each square bracket. The clauses thus obtained are in conjunctive normal form (CNF). It may be noted that p_{ij} may be in negated or non-negated form.

$$\begin{aligned} & [p_{11} \vee p_{12} \vee \dots\dots\dots \vee p_{1n}] \wedge \\ & [p_{21} \vee p_{22} \vee \dots\dots\dots \vee p_{2n}] \wedge \\ & \dots\dots\dots \\ & [p_{m1} \vee p_{m2} \vee \qquad\qquad\qquad \vee p_{mn}] \end{aligned}$$

3. Repeat

- Select any two clauses from S, such that one clause contains a negated literal and the other clause contains its corresponding positive (non-negated) literal.
- Resolve these two clauses and call the resulting clause the resolvent. Remove the parent clauses from S.

Until a null clause is obtained or no further progress can be made.

4. If a null clause is obtained, then report: "goal is proved".

Now we illustrate the general steps of resolution proof using a very simple example, which also includes the conversion of clause form.

Given: A man is a person.
 All persons will die.
 Socrates is a person.

Goal: We want to prove that Socrates will die.

The proof can be done by invoking modus-ponens twice (try it!). If we use resolution proof instead, we first do the conversion. The result of conversion for each predicate statement (left) is shown in the right.

Predicate form	Clause form
From what is given:	
1. $\forall X \text{ man}(X) \rightarrow \text{person}(X)$	$\sim \text{man}(X) \vee \text{person}(X)$
2. $\text{man}(\text{socrates})$	$\text{man}(\text{socrates})$
3. $\forall Y (\text{person}(Y) \rightarrow \text{die}(Y))$	$\sim \text{person}(Y) \vee \text{die}(Y)$
Negated goal:	
4. $\sim \text{die}(\text{socrates})$	$\sim \text{die}(\text{socrates})$

The process of the resolution proof is shown in figure 7.2. The two clauses, " $\sim \text{man}(X) \vee \text{person}(X)$ " and " $\sim \text{person}(Y) \vee \text{die}(Y)$ " are resolved to form a resolvent " $\sim \text{man}(Y) \vee \text{die}(Y)$." These two clauses can be resolved because they are opposite literals. (Since resolvents are produced by two clauses, the resolution method discussed here is called binary resolution.) The resolvent is constructed by take the disjunction of all the other atoms appearing in these two parent clauses. Also note the role of unification in this process. The entire proof is done as the construction of the resolution tree (the root is at bottom, an empty clause as the root indicates a contradiction has been found).

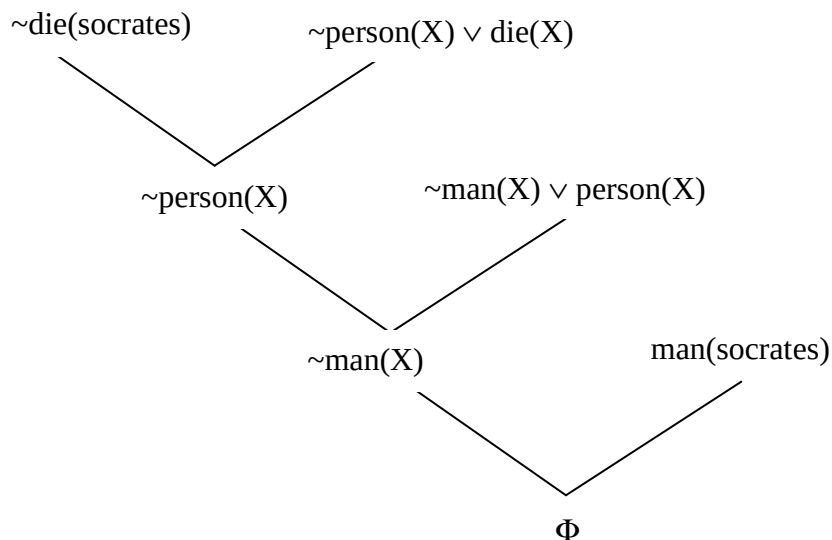


Figure 7.2: A resolution proof tree

Example 7.18: Consider the following knowledge base:

1. The humidity is high $\vee$ the sky is cloudy.
2. If the sky is cloudy then it will rain
3. If the humidity is high then it is hot.
4. It is not hot

and the goal : it will rain.

Prove by resolution theorem that the goal is derivable from the knowledge base.

Proof: Let us first denote the above clauses by the following symbols.

p = the-humidity-is-high, q = the-sky-is-cloudy, r = it-will-rain, s = it-is-hot.

The CNF form of the above clauses thus become

1. $p \vee q$
2. $\sim q \vee r$ ($q \rightarrow r$)
3. $\sim p \vee s$ ($p \rightarrow s$)
4. $\sim s$

and the negated goal = $\sim r$. Set S thus includes all these 5 clauses. Now by resolution algorithm, we construct the graph of figure 7.3. Since it terminates with a null clause, the goal is proved.

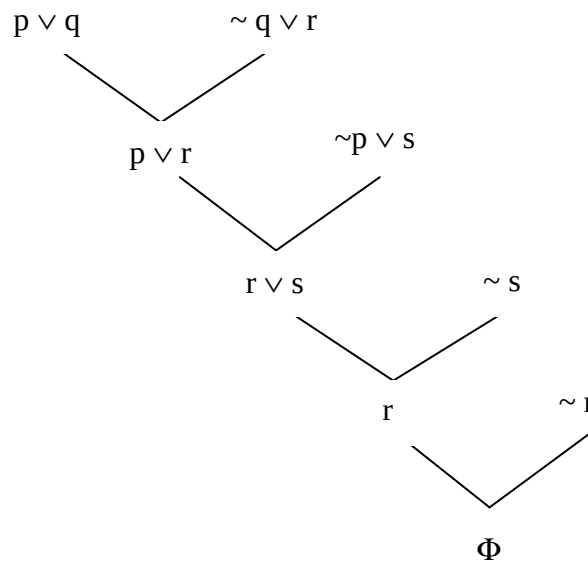
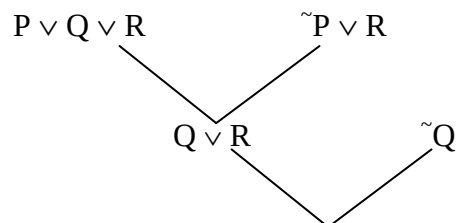


Figure 7.3: Resolution tree to prove that it-will-rain

Example 7.19: Show that o/p of resolution using following predicates

- 1) $P \vee Q \vee R$
- 2) $\sim P \vee R$
- 3) $\sim Q$
- 4) $\sim R$

Solution:



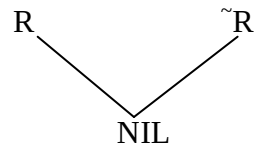


Figure 7.4: Resolution tree

Example 7.20: Everyone who see's a movie in a theater has to buy a ticket. Person who does not have money cannot buy a ticket. John see's a movie show that

- 1) John buys a ticket
- 2) John has got money
- 3) John does not have money.

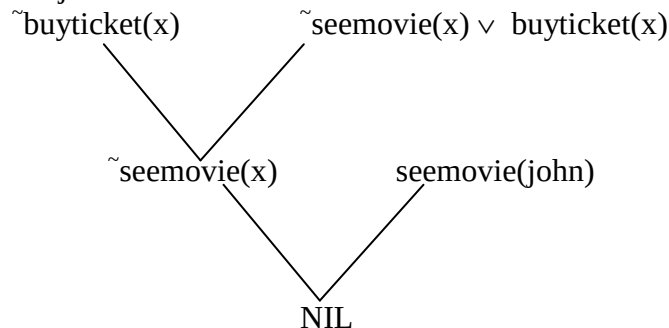
Solution: $\forall x \text{ seemovie}(x) \rightarrow \text{buyticket}(x)$
 $\forall x \sim \text{havemoney}(x) \rightarrow \sim \text{buyticket}(x)$

After conversion

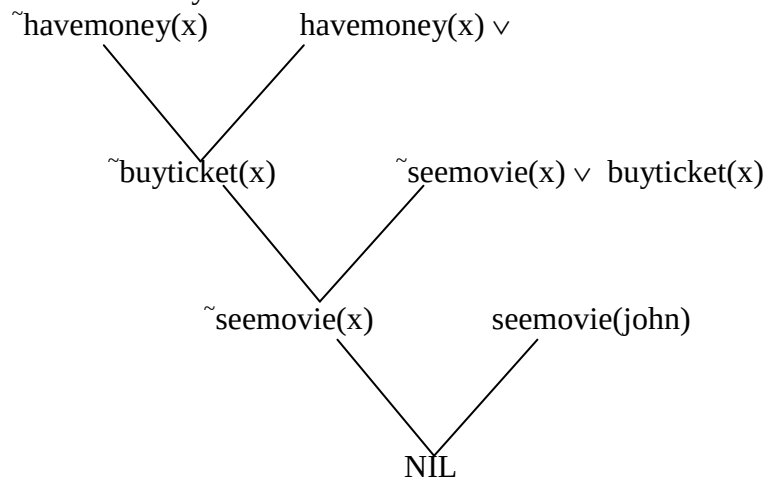
- a) $\sim \text{seemovie}(x) \vee \text{buyticket}(x)$
- b) $\text{havemoney}(x) \vee \sim \text{buyticket}(x)$
- c) $\text{seemovie}(\text{john})$

Now apply method of resolution

- 1) Not buy ticket john



- 2) John does not have money



- 3) $\text{Havemoney}(\text{john}) \rightarrow$ No negation of it, Not able to prove so he is having money.

Example 7.21: Determine whether the following argument is valid:

“If I work whole night on this problem, then I can solve it. If I solve the problem, then I will understand the topic. Therefore I will work whole night on this problem and then I will understand the topic.”

Solution: first choose propositions as

If I work whole night on this problem	:	work(problem)
I can solve the problem	:	solve(problem)
I will understand the topic	:	understand(topic)

Convert the query sentences into propositional logic

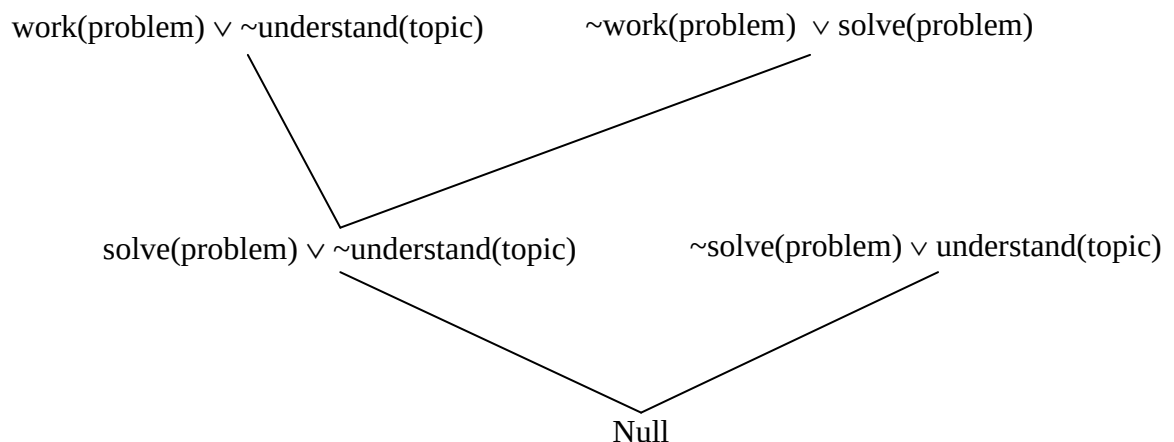
$\text{work}(\text{problem}) \rightarrow \text{solve}(\text{problem})$
 $\text{solve}(\text{problem}) \rightarrow \text{understand}(\text{topic})$
 $\text{work}(\text{problem}) \rightarrow \text{understand}(\text{topic})$

Remove implications

(a) $\sim \text{work}(\text{problem}) \vee \text{solve}(\text{problem})$
 (b) $\sim \text{solve}(\text{problem}) \vee \text{understand}(\text{topic})$
 (c) $\sim \text{work}(\text{problem}) \vee \text{understand}(\text{topic})$

Make negation of goal and proof by refutation.

$\text{work}(\text{problem}) \vee \sim \text{understand}(\text{topic})$



As we get null clause i.e. contradiction hence arguments are valid.

Example 7.22: Formulate FOPL of the following statements.

- 1) Marcus was a man
- 2) Marcus was a Pompeian's
- 3) All Pompeian's were Romans.
- 4) Caesar was a ruler.
- 5) All Romans were either loyal to Caesar or hated him.
- 6) Everyone is loyal to someone.
- 7) People only try to assassinate rulers they are not loyal to.
- 8) Marcus tried to assassinate Caesar.

Solution:

- 1) man (Marcus)
- 2) Pompeian (Marcus)
- 3) $\forall x \text{ Pompeian}(x) \rightarrow \text{Roman}(x)$
- 4) ruler (Caesar)

- 5) $\forall x \text{ Roman}(x) \rightarrow \text{loyalto}(x, \text{Caesar}) \vee \text{hated}(x, \text{Caesar})$
- 6) $\forall x \exists y \text{ loyalto}(x, y)$
- 7) $\forall x \forall y \text{ Person}(x) \wedge \text{ruler}(y) \wedge \text{trytoassassinate}(x, y) \rightarrow \sim \text{loyalto}(x, y)$
- 8) $\text{trytoassassinate}(\text{Marcus}, \text{Caesar})$.

Suppose we want to use above statements to answer.

WAS MARCUS LOYAL TO CAESAR?

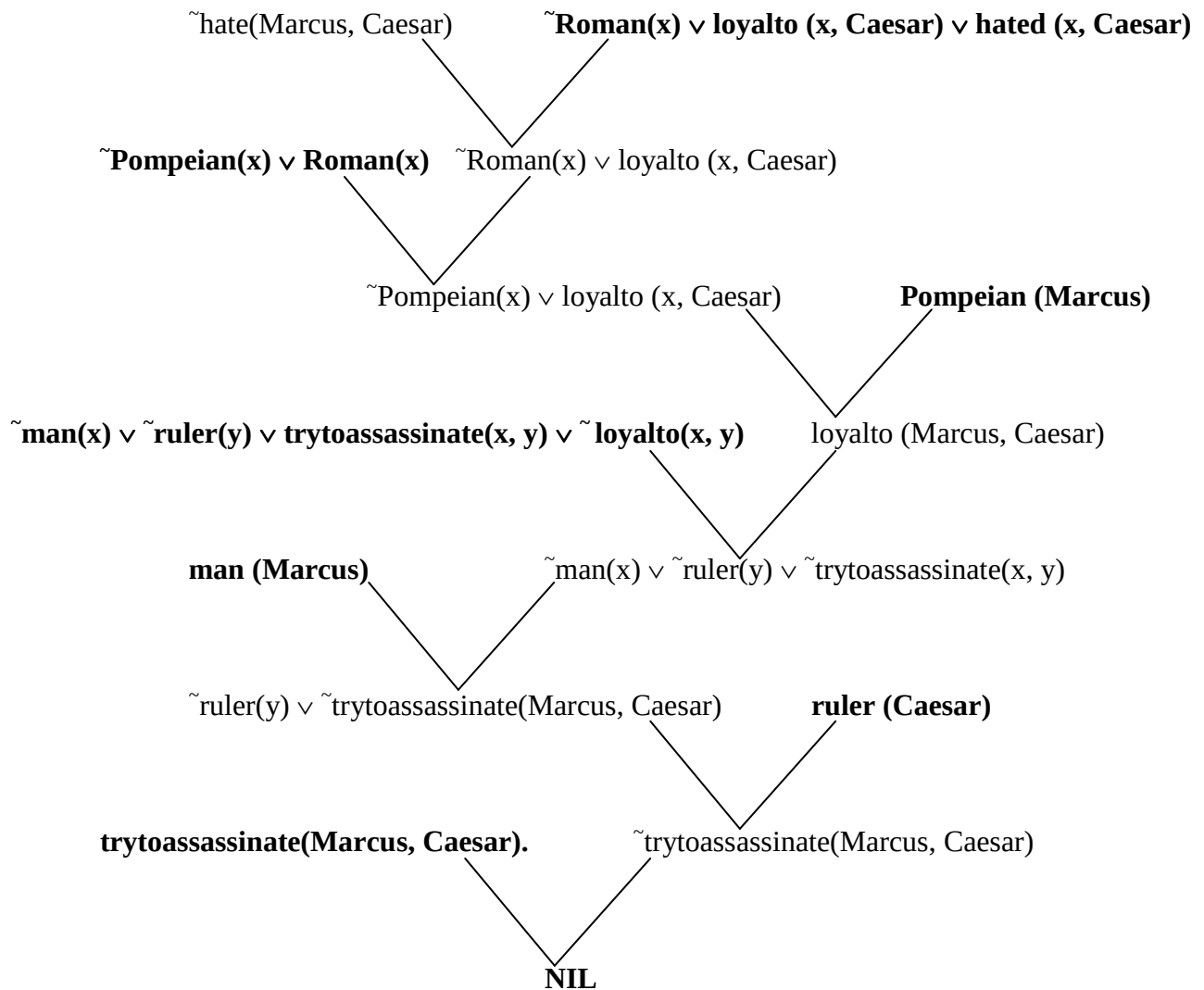
$$\begin{array}{l} \sim \text{loyalto}(\text{Marcus}, \text{Caesar}) \\ \uparrow (7, \text{substitution}) \\ \text{Person}(\text{Marcus}) \wedge \text{Ruler}(\text{Caesar}) \wedge \text{trytoassassinate}(\text{Marcus}, \text{Caesar}) \\ \uparrow (4, \text{substitution}) \\ \text{Person}(\text{Marcus}) \wedge \text{trytoassassinate}(\text{Marcus}, \text{Caesar}) \\ \uparrow (8, \text{substitution}) \\ \text{Person}(\text{Marcus}) \end{array}$$

An attempt to prove $\sim \text{loyalto}(\text{Marcus}, \text{Caesar})$, Marcus was not loyal to Caesar.

Example 7.23: Using statements below answer “WAS MARCUS LOYAL TO CAESAR?”.

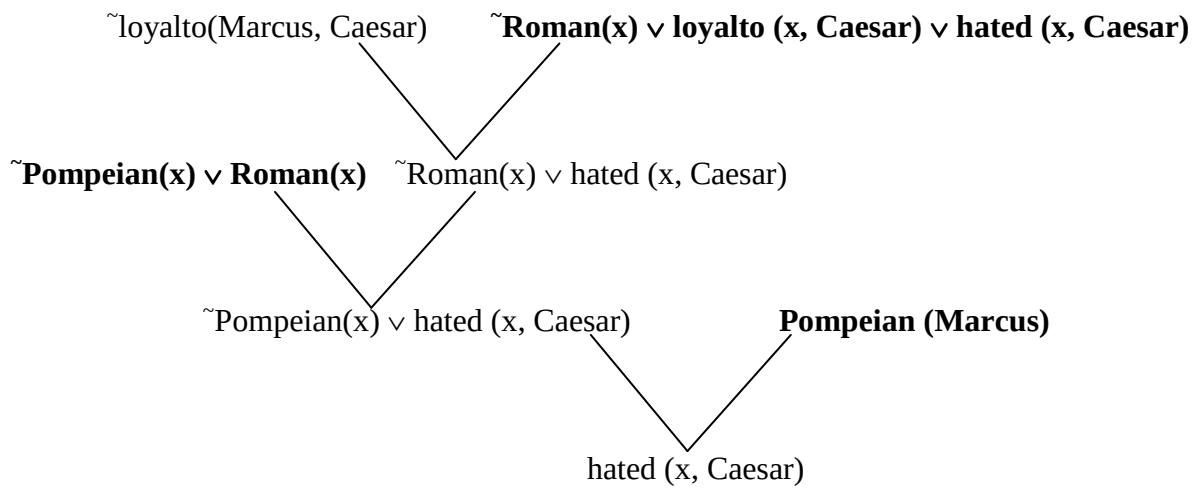
- 1) $\text{man}(\text{Marcus})$
- 2) $\text{Pompeian}(\text{Marcus})$
- 3) $\sim \text{Pompeian}(x) \vee \text{Roman}(x)$
- 4) $\text{ruler}(\text{Caesar})$
- 5) $\sim \text{Roman}(x) \vee \text{loyalto}(x, \text{Caesar}) \vee \text{hated}(x, \text{Caesar})$
- 6) $\text{loyalto}(x, y)$
- 7) $\sim \text{man}(x) \vee \sim \text{ruler}(y) \vee \sim \text{trytoassassinate}(x, y) \vee \sim \text{loyalto}(x, y)$
- 8) $\text{trytoassassinate}(\text{Marcus}, \text{Caesar})$.

Prove: $\text{Hate}(\text{Marcus}, \text{Caesar})$



Hence proved Marcus was not loyal to Caesar

Prove: Loyalto (Marcus, Caesar)



Hence proved Marcus was not loyal to Caesar